

PY1010 — Physics for Biological and Chemical Sciences

Summer 2022 Examination · Paper 1 · Full Marking Scheme

Module	PY1010 — Physics for Biological and Chemical Sciences
Internal examiners	Dr John Cuffe, Prof. Paul Callanan
Institution	University College Cork
Session	Summer 2022, Paper 1
Duration	3 hours (+ 15 minutes reading time)
Structure	Six questions, each given as an either/or choice (A or B). Answer one option of each of the six questions and all parts of that option. Marks for each part are shown in square brackets.
Materials	Non-programmable calculator permitted; constants on the paper's "useful data" page

Coverage note. This scheme works through **both options (A and B) of all six questions** in full. On the day you only choose one option per question, but working both lets you pick the one you're strongest on.

Source basis. Every answer is built **only** from the PY1010 lecture material given to students: the lecture slides (mechanics, fluids, thermodynamics) and the handwritten lecture notes (gas laws, electricity, magnetism, waves and optics). No textbook or outside material is used, except where a highlighted note says the lecture notes don't cover something (Q5(B)(c) and Q6(A)(a)).

Own-figure rule. Where a part carries forward a number from an earlier part, an answer that uses the correct method on an earlier wrong value still earns the method marks for the later part.

Numerical conventions. $g = 9.81 \text{ m s}^{-2}$ and other constants from the paper's useful-data page. Intermediate values are kept unrounded, and final answers are given to 2–3 significant figures. Answers differing only through rounding should receive full marks.

Study-Uni

Question 1 · Option A

Order-of-magnitude estimates: grains of sand vs stars in the universe — 10 marks

Given: 30% of the world's coastline is sandy; each grain of sand occupies 1 mm^3 .

1(A)(a)

[1 mark]

If the total length of coastline on Earth is ~ 1 million km, estimate the total length of sandy beaches, in km.

ANSWER

$\approx 3 \times 10^5 \text{ km of sandy beach}$

WORKING

$$L = 0.30 \times 10^6 \text{ km} = 3 \times 10^5 \text{ km}$$

Marks

Awarded for

1

$3 \times 10^5 \text{ km.}$

1(A)(b)**[3 marks]**

If the beaches span 50 m from land to sea and are 25 m deep, what is the total volume of sand in all the beaches in the world, in m³?

ANSWER

$$V \approx 3.8 \times 10^{11} \text{ m}^3$$

WORKING

Convert the length to metres: $3 \times 10^5 \text{ km} = 3 \times 10^8 \text{ m}$. Treat the beaches as a long box: volume = length $\times$ width $\times$ depth.

$$V = 3 \times 10^8 \text{ m} \times 50 \text{ m} \times 25 \text{ m} = 3.75 \times 10^{11} \text{ m}^3$$

Marks	Awarded for
1	Length converted to metres.
1	$V = \text{length} \times \text{width} \times \text{depth}$.
1	$\approx 3.8 \times 10^{11} \text{ m}^3$.

1(A)(c)**[3 marks]**

Hence, estimate how many grains of sand there are in all the beaches in the world.

ANSWER

$N \approx 4 \times 10^{20}$ grains of sand

WORKING

Volume of one grain: $1 \text{ mm}^3 = (10^{-3} \text{ m})^3 = 10^{-9} \text{ m}^3$.

$$N = 3.75 \times 10^{11} \text{ m}^3 \div 10^{-9} \text{ m}^3 = 3.75 \times 10^{20} \text{ grains}$$

EXAM TECHNIQUE

The cube of the conversion factor is the key step: $1 \text{ mm} = 10^{-3} \text{ m}$, so $1 \text{ mm}^3 = 10^{-9} \text{ m}^3$, not 10^{-3} m^3 .

Marks	Awarded for
1	$1 \text{ mm}^3 = 10^{-9} \text{ m}^3$.
1	$N = V_{\text{total}}/V_{\text{grain}}$.
1	$\approx 4 \times 10^{20}$.

1(A)(d)**[3 marks]**

There are $\sim 10^{11}$ galaxies, each with $\sim 10^{11}$ stars like the Milky Way. What is the total number of stars in the universe? How does it compare to the number of grains of sand from (c)?

ANSWER **10^{22} stars: about 25-30 times more stars than grains of sand****WORKING**

$$N_{\text{stars}} = 10^{11} \times 10^{11} = 10^{22}$$

$$N_{\text{stars}} / N_{\text{sand}} = 10^{22} \div 3.75 \times 10^{20} \approx 27$$

Marks	Awarded for
1	10^{22} stars.
1	Ratio calculated.
1	Conclusion: more stars than grains of sand (own-figure from (c) accepted).

Question 1(A) Total: 10 Marks

Question 1 · Option B

Speed vs velocity, kinematic equations, road bowling — 10 marks

1(B)(a)

[2 marks]

Define and give the SI units for (i) speed (ii) velocity.

ANSWER

Speed = distance/time (scalar); velocity = displacement/time (vector); both in m s^{-1}

WORKING

(i) **Speed** = total distance ÷ total time. It has magnitude only (a scalar). Unit: m s^{-1} .

(ii) **Velocity** = displacement ÷ time, $v = \Delta x / \Delta t$. It has magnitude **and direction** (a vector). Unit: m s^{-1} .

Marks	Awarded for
1	Speed defined, m s^{-1} .
1	Velocity as a vector (displacement/time), m s^{-1} .

1(B)(b)

[1 mark]

Under what condition are the Kinematic Equations of Motion valid?

ANSWER

Only when the acceleration is constant

WORKING

The equations ($v = v_0 + at$, $x = x_0 + v_0 t + \frac{1}{2}at^2$, $v^2 = v_0^2 + 2a(x - x_0)$) only hold when the acceleration is **constant**.

Marks	Awarded for
1	Constant acceleration.

1(B)(c)(i)**[3 marks]**

A road-bowling ball travels 60 m in 5 s before rolling to a halt, decelerating at a constant rate. Calculate its initial velocity.

ANSWER

$$v_0 = 24 \text{ m s}^{-1}$$

WORKING

Constant deceleration, so use $x = \frac{1}{2}(v + v_0)t$ with final velocity $v = 0$:

$$60 = \frac{1}{2} (0 + v_0) \times 5 \Rightarrow v_0 = 2 \times 60 \div 5 = 24 \text{ m s}^{-1}$$

Marks	Awarded for
1	$v = 0$ at the end; suitable equation chosen.
1	Correct substitution.
1	24 m s^{-1} .

1(B)(c)(ii)**[4 marks]**

He throws the ball vertically upwards at the velocity from (i), standing next to a 28 m high bridge. What maximum height will it reach? Is it possible for it to pass over the bridge?

ANSWER

$h \approx 29.4$ m: just possible, but only by ≈ 1.4 m

WORKING

Going up, $a = -g$ and at the top $v = 0$. Use $v^2 = v_0^2 + 2a(x - x_0)$:

$$0 = 24^2 - 2 \times 9.81 \times h \Rightarrow h = 576 \div 19.62 = 29.4 \text{ m}$$

29.4 m is greater than 28 m, so the ball could just clear the bridge. The margin is only about 1.4 m, and air resistance (ignored here) would reduce the height, so in practice it would be very close.

Marks	Awarded for
1	$a = -g$ and $v = 0$ at the top.
1	$v^2 = v_0^2 + 2ah$ rearranged.
1	$h \approx 29.4$ m (own-figure accepted).
1	Compares with 28 m and concludes (just) possible.

Question 1(B) Total: 10 Marks

Question 2 · Option A

Power, energy cost of accelerating a car, driving speed and fuel use — 10 marks

Given: Car of mass 1000 kg, $0 \rightarrow 100$ km/hr in 10 s. Petrol €2.00 per litre, 10^7 J per litre.
Electricity 30 c per kW hr.

2(A)(a)

[2 marks]

Define power, and give its SI unit.

ANSWER

$P = W/t$; unit the watt, $W = J\ s^{-1}$

WORKING

Power is the rate of doing work (or of releasing energy): $P = W / t$.

Marks	Awarded for
1	Rate of doing work, $P = W/t$.
1	Watt ($J\ s^{-1}$).

2(A)(b)(i)**[2 marks]**

If petrol costs €2.00 per litre, and each litre gives 10^7 J to the engine, how much does it cost to accelerate the car to 100 km/hr (in cent)?

ANSWER

≈ 7.7 cent (assuming all the fuel energy becomes kinetic energy)

WORKING

Speed: $v = 100 \text{ km/hr} = 100\,000 \text{ m} \div 3600 \text{ s} = 27.8 \text{ m s}^{-1}$. The work done by the engine equals the car's gain in kinetic energy:

$$E_{\text{kin}} = \frac{1}{2} m v^2 = \frac{1}{2} \times 1000 \times (27.78)^2 = 3.86 \times 10^5 \text{ J}$$

$$\text{Petrol used} = 3.86 \times 10^5 \text{ J} \div 10^7 \text{ J/L} = 0.0386 \text{ L; cost} = 0.0386 \times 200 \text{ c} = 7.7 \text{ c}$$

Marks	Awarded for
1	$E_{\text{kin}} = \frac{1}{2}mv^2$ with v in m s^{-1} .
1	$\approx 7.7 \text{ c.}$

2(A)(b)(ii)**[2 marks]**

If the engine were electric, how much would it cost instead, at 30 c per kW hr?

ANSWER

≈ 3.2 cent

WORKING

1 kW hr is 1000 W for 3600 s, so $1 \text{ kW hr} = 1000 \times 3600 = 3.6 \times 10^6 \text{ J}$.

Energy = $3.86 \times 10^5 \text{ J} \div 3.6 \times 10^6 \text{ J per kW hr} = 0.107 \text{ kW hr}$

Cost = $0.107 \times 30 \text{ c} = 3.2 \text{ c}$

Marks	Awarded for
1	1 kW hr = $3.6 \times 10^6 \text{ J}$.
1	$\approx 3.2 \text{ c}$.

2(A)(c)(i)**[2 marks]**

Car A travels 100 km at 80 km/hr and car B at 100 km/hr. If the power used depends on the square of the velocity, how much more power does car B use, relative to car A?

ANSWER

About 1.56 times the power ($\approx 56\%$ more)

WORKING

$$P_B / P_A = (v_B / v_A)^2 = (100/80)^2 = 1.5625$$

Marks	Awarded for
1	$P \propto v^2$, ratio $(100/80)^2$.
1	1.56 (56% more).

2(A)(c)(ii)**[2 marks]**

How much more energy does car B use, relative to car A?

ANSWER**1.25 times the energy (25% more)****WORKING**

Energy = power × time, and the time to cover 100 km is $t = d/v$. Car B takes only $80/100 = 0.8$ of car A's time:

$$E_B / E_A = (P_B / P_A) \times (t_B / t_A) = 1.5625 \times 0.8 = 1.25$$

Marks	Awarded for
1	$E = Pt$ with $t = d/v$.
1	1.25 (25% more).

Question 2(A) Total: 10 Marks

Question 2 · Option B

Centripetal force and acceleration, the International Space Station — 10 marks

Given: ISS orbits 390 km above the surface; mass 4.5×10^5 kg; take g the same as at the Earth's surface; Earth radius 6400 km.

2(B)(a)

[1 mark]

Explain what is meant by centripetal force.

ANSWER

The net force, directed towards the centre, needed to keep an object moving in a circle

WORKING

In uniform circular motion the direction of the velocity changes constantly, so the object is always accelerating towards the centre of the circle. The centripetal force is the **net force directed towards the centre** that causes this acceleration and keeps the object moving in a circle.

Marks

Awarded for

1

Net force towards the centre that keeps the object in circular motion.

2(B)(b)

[1 mark]

The centripetal force is mv^2/r . From this, write down an expression for the acceleration due to this force.

ANSWER

$a = v^2/r$ (directed towards the centre)

WORKING

From $F = ma$:

Marks

Awarded for

1

$a = v^2/r$.

2(B)(c)(i)**[1 mark]**

What thrust is required from its engines to keep the ISS in orbit?

ANSWER

Zero: gravity alone supplies the centripetal force

WORKING

None. Gravity provides the centripetal force that bends its path into a circle. No force is needed along the direction of motion, since by Newton's first law it keeps moving at constant speed on its own (ignoring any air drag).

Marks	Awarded for
1	Zero thrust, with reason.

2(B)(c)(ii)**[2 marks]**

What gravitational force (in N) acts on the ISS?

ANSWER

$F \approx 4.4 \times 10^6 \text{ N}$

WORKING

$$F = m g = 4.5 \times 10^5 \text{ kg} \times 9.81 \text{ m s}^{-2} = 4.4 \times 10^6 \text{ N}$$

Marks	Awarded for
1	$F = mg$ ($w = mg$).
1	$4.4 \times 10^6 \text{ N}$.

2(B)(c)(iii)**[2 marks]**

What is (A) the magnitude and (B) the direction of the centripetal force acting on the ISS?

ANSWER

(A) $4.4 \times 10^6 \text{ N}$; (B) towards the centre of the Earth

WORKING

The centripetal force **is** the gravitational force.

Marks	Awarded for
1	Magnitude $4.4 \times 10^6 \text{ N}$.
1	Direction: towards the Earth's centre.

2(B)(c)(iv)**[2 marks]**

Calculate the speed (in m s^{-1}) at which the ISS travels as it orbits the Earth.

ANSWER

$v \approx 8.2 \times 10^3 \text{ m s}^{-1}$ (8.2 km s^{-1})

WORKING

The orbit radius is measured from the Earth's centre: $r = 6400 + 390 = 6790 \text{ km} = 6.79 \times 10^6 \text{ m}$.

Set gravity equal to the centripetal force: $mg = mv^2/r$, so $v = \sqrt{(g r)}$:

$$v = \sqrt{(9.81 \times 6.79 \times 10^6)} = 8.2 \times 10^3 \text{ m s}^{-1}$$

Marks	Awarded for
1	$r = R_E + 390 \text{ km}$ and $mg = mv^2/r$.
1	$\approx 8.2 \times 10^3 \text{ m s}^{-1}$.

2(B)(c)(v)**[1 mark]**

If, while the ISS were orbiting, the centripetal force were somehow switched off, what would happen to it?

ANSWER

It would fly off in a straight line along the tangent to its orbit

WORKING

With no net force, by Newton's first law it would continue at constant velocity in a **straight line**, along the tangent to its orbit at that moment, and fly off into space.

Marks	Awarded for
-------	-------------

1	Straight line / tangent (Newton's first law).
---	---

Question 2(B) Total: 10 Marks

Question 3 · Option A

Archimedes' principle, a helium weather balloon — 10 marks

Given: Spherical balloon, radius 2 m; $\rho_{\text{He}} = 0.18 \text{ kg m}^{-3}$; $\rho_{\text{air}} = 1.2 \text{ kg m}^{-3}$; other components' mass negligible.

3(A)(a)

[2 marks]

State Archimedes' Principle.

ANSWER

Buoyant force = weight of fluid displaced

WORKING

When a body is completely or partially immersed in a fluid, the fluid exerts an **upward force** on the body equal to the **weight of the fluid displaced** by the body. The principle also holds for gases such as air.

Marks	Awarded for
1	Upward force on an immersed body.
1	Equal to the weight of fluid displaced.

3(A)(b)(i)**[2 marks]**

What upward, buoyant force does the balloon experience?

ANSWER

$F_{\text{buoy}} \approx 394 \text{ N (upwards)}$

WORKINGBalloon volume: $V = (4/3)\pi(2 \text{ m})^3 = 33.5 \text{ m}^3$. The balloon displaces this volume of air:

$$F_{\text{buoy}} = \rho_{\text{air}} V g = 1.2 \times 33.5 \times 9.81 = 394 \text{ N}$$

Marks	Awarded for
1	$V = (4/3)\pi r^3$ and $F = \rho_{\text{air}} V g$.
1	$\approx 394 \text{ N}$.

3(A)(b)(ii)**[3 marks]**

What will the initial acceleration of the balloon be? Give both magnitude and direction. Ignore air resistance.

ANSWER

$a \approx 56 \text{ m s}^{-2}$, vertically upwards

WORKING

Mass of helium: $m = \rho_{\text{He}} V = 0.18 \times 33.5 = 6.03 \text{ kg}$, so its weight is $W = mg = 59.2 \text{ N}$.

Net upward force: $F_{\text{net}} = 394 - 59.2 = 335 \text{ N}$. Newton's second law:

$$a = F_{\text{net}} / m = 335 \text{ N} \div 6.03 \text{ kg} = 55.6 \text{ m s}^{-2}$$

Marks	Awarded for
1	Weight of the helium (the only mass).
1	Net force = buoyant force – weight, $a = F/m$.
1	$\approx 56 \text{ m s}^{-2}$ upwards.

3(A)(b)(iii)**[3 marks]**

As the balloon rises, the air pressure decreases. What effect does this have on the volume of the balloon, and the density of the gas within it?

ANSWER

The volume increases (Boyle's law, $P \propto 1/V$) and the density of the helium decreases (same mass, larger volume)

WORKING

The helium inside is at the same pressure as the surrounding air. For a fixed amount of gas at roughly constant temperature, Boyle's law gives $PV = \text{constant}$, so as P falls, **V increases** and the balloon expands. This is like the rising bubble example in the lectures.

The mass of helium stays the same while the volume increases, so the **density** of the gas ($\rho = m/V$) **decreases**.

Marks	Awarded for
1	Volume increases.
1	Explained via Boyle's law ($PV = \text{constant}$).
1	Density decreases ($\rho = m/V$, m fixed).

Question 3(A) Total: 10 Marks

Question 3 · Option B

Pascal's principle, pressure at depth, a hydraulic lift — 10 marks

3(B)(a)

[2 marks]

State Pascal's Principle, and give the SI unit for pressure.

ANSWER

Pressure is transmitted undiminished throughout an enclosed fluid; unit the pascal, $\text{Pa} = \text{N m}^{-2}$

WORKING

The pressure applied at one point in an enclosed fluid is transmitted **undiminished** to every part of the fluid and to the walls of the container.

Marks	Awarded for
1	Pascal's principle.
1	Pa (N m^{-2}).

3(B)(b)(i)

[1 mark]

A diver is 50 m below the surface. Does the diver experience pressure (a) mostly from above, (b) mostly from below, or (c) equally in all directions?

ANSWER

(c) Equally in all directions

WORKING

At a given depth the pressure exerted by a fluid is the same in all directions.

Marks	Awarded for
1	(c).

3(B)(b)(ii)**[3 marks]**

To breathe safely, air must be supplied to the diver at ambient pressure.
Calculate this pressure.

ANSWER

$P \approx 5.9 \times 10^5 \text{ Pa}$ (about 6 atmospheres)

WORKING

The pressure at depth h is atmospheric pressure plus the pressure of the water above:

$$P = P_{\text{atm}} + \rho g h = 1.013 \times 10^5 + (1000 \times 9.81 \times 50)$$

$$P = 1.013 \times 10^5 + 4.905 \times 10^5 = 5.9 \times 10^5 \text{ Pa}$$

Marks	Awarded for
1	$P = P_{\text{atm}} + \rho g h$.
1	$\rho g h = 4.9 \times 10^5 \text{ Pa}$.
1	$\approx 5.9 \times 10^5 \text{ Pa}$.

3(B)(c)**[4 marks]**

A clown (weight 800 N) stands on a water-filled piston of area 2 cm², connected to a similar piston of area 60 cm² on which an elephant stands. Use Pascal's Principle to calculate the maximum mass of the elephant (in kg) that these pistons can lift.

ANSWER

$$m_{\max} \approx 2.4 \times 10^3 \text{ kg (about 2450 kg)}$$

WORKING

By Pascal's principle the pressure is the same on both pistons:

$$F_{\text{clown}} / A_{\text{clown}} = F_{\text{elephant}} / A_{\text{elephant}}$$

$$F_{\text{elephant}} = 800 \text{ N} \times (60 \text{ cm}^2 / 2 \text{ cm}^2) = 24\,000 \text{ N}$$

The areas appear as a ratio, so the units cancel and there's no need to convert to m². The elephant's weight is at most 24 000 N:

$$m = 24\,000 \div 9.81 = 2.4 \times 10^3 \text{ kg}$$

Marks	Awarded for
1	Equal pressures on the two pistons.
1	$F_{\text{elephant}} = F_{\text{clown}} \times A \text{ ratio.}$
1	24 000 N.
1	$m = F/g \approx 2450 \text{ kg.}$

Question 3(B) Total: 10 Marks

Question 4 · Option A

Coulomb's law, forces between charges, equilibrium position, field lines — 10 marks

Given: $q_1 = +10 \mu\text{C}$ and $q_2 = +90 \mu\text{C}$, fixed at the left and right ends of a 1 m stick.

4(A)(a)

[2 marks]

State Coulomb's law for the magnitude of the force F between two point charges q_1 and q_2 a distance r apart. State the SI units of all quantities. When would F be repulsive or attractive?

ANSWER

$F = q_1 q_2 / (4\pi\epsilon_0 r^2)$; repulsive for like charges, attractive for unlike

WORKING

$$F = q_1 q_2 / (4\pi\epsilon_0 r^2)$$

F in N; q_1, q_2 in C; r in m; ϵ_0 in $\text{C}^2 \text{J}^{-1} \text{m}^{-1}$ (the constant $1/(4\pi\epsilon_0) = 8.99 \times 10^9 \text{ N m}^2 \text{C}^{-2}$).

Like charges (same sign) **repel**; unlike charges (opposite sign) **attract**.

Marks	Awarded for
1	Law with SI units.
1	Like repel, unlike attract.

4(A)(b)(i)**[2 marks]**

Calculate the force on q_1 due to q_2 and on q_2 due to q_1 , and give the direction of these forces.

ANSWER

8.1 N on each: q_1 is pushed left (away from q_2), q_2 is pushed right (away from q_1)

WORKING

$$F = 8.99 \times 10^9 \times (10 \times 10^{-6})(90 \times 10^{-6}) \div (1 \text{ m})^2 = 8.1 \text{ N}$$

Both charges are positive, so the force is repulsive. By Newton's third law the two forces are equal and opposite.

Marks	Awarded for
1	$F \approx 8.1 \text{ N}$.
1	Repulsive, equal and opposite directions.

4(A)(b)(ii)**[4 marks]**

At what point along the metre stick should a third charge be placed such that it does not move?

ANSWER

0.25 m from q_1 (0.75 m from q_2)

WORKING

Place q_3 a distance x from q_1 , so it is $(1 - x)$ from q_2 . It stays still if the two forces on it are equal and opposite, which can only happen between the charges:

$$k q_1 q_3 / x^2 = k q_2 q_3 / (1 - x)^2$$

k and q_3 cancel, so the answer doesn't depend on q_3 :

$$(1 - x)^2 / x^2 = q_2 / q_1 = 9 \Rightarrow (1 - x) / x = 3 \Rightarrow x = 0.25 \text{ m}$$

Marks	Awarded for
1	Forces on q_3 must balance, between the charges.
1	Equation set up with x and $(1 - x)$.
1	Solved: $(1 - x) / x = 3$.
1	$x = 0.25 \text{ m}$ from q_1 .

4(A)(c)**[2 marks]**

Assuming q_3 is a negative charge, draw a diagram showing the electric field lines between the charges and their direction.

ANSWER

Lines leave q_1 and q_2 (positive) and end on q_3 (negative)

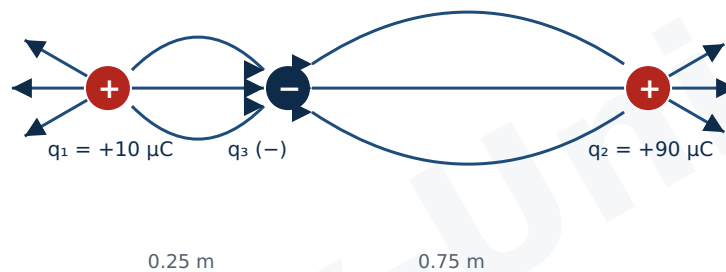
WORKING

Fig. 1 — Field lines start on the positive charges and end on the negative charge q_3 ; they also point outward from q_1 and q_2 .

E points in the direction of the force on a positive test charge. So the field lines point **away from** the positive charges q_1 and q_2 and **into** the negative charge q_3 .

Marks	Awarded for
1	Lines from q_1 and q_2 ending on q_3 .
1	Correct arrow directions.

Question 4(A) Total: 10 Marks

Question 4 · Option B

Electric field and potential; a parallel-plate capacitor — 10 marks

Given: Plates $30\text{ cm} \times 50\text{ cm}$, separated by $d = 2\text{ mm}$, connected to a 3.7 V battery.

4(B)(a)

[2 marks]

State and explain briefly the connection between the electric field and the electric potential, and give the SI units for each.

ANSWER

$V = Ed$; E in V m^{-1} , V in volts (V)

WORKING

The potential difference V between two points is the work done per unit charge in moving a charge between them, $V = W/q$. In a uniform field, moving a distance d along E gives:

$$V = E d \Leftrightarrow E = V / d$$

So the field is the potential difference per unit distance.

Marks	Awarded for
1	$V = Ed$ (or $E = V/d$) with explanation.
1	Units: V m^{-1} and V.

4(B)(b)(i)**[2 marks]**

What is the magnitude of the electric field at the mid-point between the two plates?

ANSWER

$$E = 1.85 \times 10^3 \text{ V m}^{-1}$$

WORKING

The field between the plates is uniform, so it's the same at the mid-point:

$$E = V / d = 3.7 \text{ V} \div 0.002 \text{ m} = 1850 \text{ V m}^{-1}$$

Marks	Awarded for
1	$E = V/d$ with d in m.
1	1850 V m ⁻¹ .

4(B)(b)(ii)**[2 marks]**

What is the total capacitance of the capacitor?

ANSWER

$$C \approx 6.6 \times 10^{-10} \text{ F (0.66 nF)}$$

WORKING

Plate area: $A = 0.30 \text{ m} \times 0.50 \text{ m} = 0.15 \text{ m}^2$.

$$C = \epsilon_0 A / d = 8.854 \times 10^{-12} \times 0.15 \div 0.002 = 6.6 \times 10^{-10} \text{ F}$$

Marks	Awarded for
1	$C = \epsilon_0 A/d$ with A in m ² .
1	$\approx 6.6 \times 10^{-10} \text{ F}$.

4(B)(b)(iii)**[2 marks]**

A charge of $+1 \mu\text{C}$ moves from the positive to the negative plate. What is the change in electric potential energy for this charge?

ANSWER**PE decreases by $3.7 \times 10^{-6} \text{ J}$** **WORKING**

$$\Delta U = qV = 1 \times 10^{-6} \text{ C} \times 3.7 \text{ V} = 3.7 \times 10^{-6} \text{ J}$$

A positive charge moving from the positive to the negative plate goes from higher to lower potential, so its potential energy **decreases**.

Marks	Awarded for
1	$\Delta U = qV = 3.7 \times 10^{-6} \text{ J}$.
1	Stated as a decrease.

4(B)(b)(iv)**[2 marks]**

If this energy is converted to kinetic energy, and the charge has a mass of 1.87 mg, what is its final velocity? (It started from rest.)

ANSWER

$$v \approx 2.0 \text{ m s}^{-1}$$

WORKING

Mass: $1.87 \text{ mg} = 1.87 \times 10^{-6} \text{ kg}$. Setting $\frac{1}{2}mv^2 = \Delta U$:

$$v = \sqrt{(2 \times 3.7 \times 10^{-6} \div 1.87 \times 10^{-6})} = \sqrt{3.96} = 2.0 \text{ m s}^{-1}$$

Marks	Awarded for
1	$\frac{1}{2}mv^2 = \Delta U$ with mass in kg.
1	$\approx 2.0 \text{ m s}^{-1}$.

Question 4(B) Total: 10 Marks

Question 5 · Option A

Lorentz force on a moving charge; force between parallel wires — 10 marks

5(A)(a)

[2 marks]

Write down an expression for F , the Lorentz force on a charge q moving at speed v at an angle θ to a magnetic field B . State the SI units of all quantities.

ANSWER

$$F = qvB \sin \theta; F \text{ (N)}, q \text{ (C)}, v \text{ (m s}^{-1}\text{)}, B \text{ (T)}$$

WORKING

$$F = q v B \sin \theta$$

F in N, q in C, v in m s^{-1} , B in T. When $v \perp B$ ($\theta = 90^\circ$) this becomes $F = qvB$, the maximum. When v is parallel to B ($\theta = 0$) there is no force.

Note on the lecture material: The lecture notes give the Lorentz force for $v \perp B$ only ($F = BQv$) and note that a velocity component along B produces no force (helical motion). The general form with $\sin \theta$ goes slightly beyond this.

Marks	Awarded for
1	$F = qvB \sin \theta$.
1	All SI units.

5(A)(b)**[2 marks]**

A positive charge $q = +10 \mu\text{C}$ moves at 800 m/s perpendicular to a field of 1.2 T . What is the magnitude of the Lorentz force on q ?

ANSWER

$F = 9.6 \times 10^{-3} \text{ N}$

WORKING

$$F = q v B = 10 \times 10^{-6} \text{ C} \times 800 \text{ m s}^{-1} \times 1.2 \text{ T} = 9.6 \times 10^{-3} \text{ N}$$

Marks	Awarded for
1	$F = qvB$ with μC converted.
1	$9.6 \times 10^{-3} \text{ N}$.

5(A)(c)**[2 marks]**

A negative charge q travels in the x -direction. A magnetic field B pointing in the y -direction (into the page) is switched on. What is the direction of the force on q ?

ANSWER

Along $-z$ (down the page)

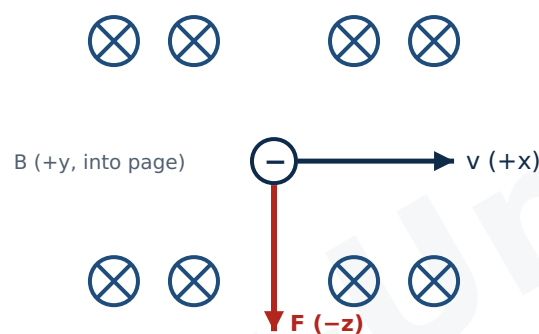
WORKING

Fig. 2 — A negative charge moving along $+x$ through B into the page is pushed down, along $-z$.

A negative charge moving along $+x$ is equivalent to a conventional current along $-x$.

Left Hand Rule with B into the page and current to the left: the force points **down the page**, along $-z$. For a positive charge the force would be along $+z$, and the negative sign reverses it.

Marks	Awarded for
1	Uses current direction opposite to the electron's motion (or reverses the positive-charge result).
1	$-z$.

5(A)(d)(i)**[2 marks]**

Two parallel wires, each 4 m long, carry the same current and are 0.1 m apart in a vacuum. The force on each is 2×10^{-4} N. What is the current I in each wire?

ANSWER

$$I = 5 \text{ A}$$

WORKING

Force per unit length between two wires with equal currents:

$$F / \ell = (\mu_0 / 2\pi) \cdot I^2 / d$$

$$I^2 = (F/\ell) \times d \div (\mu_0/2\pi) = (2 \times 10^{-4} / 4) \times 0.1 \div (2 \times 10^{-7}) = 25 \Rightarrow I = 5 \text{ A}$$

Marks	Awarded for
1	Correct formula $F/\ell = \mu_0 I^2 / (2\pi d)$.
1	$I = 5 \text{ A}$.

5(A)(d)(ii)**[2 marks]**

If the current in each wire is tripled, by what factor does the force change?

ANSWER**The force increases by a factor of 9****WORKING**

$F \propto I^2$ (one factor of I from each wire). Tripling both currents:

$$F_{\text{new}} / F = 3^2 = 9$$

Marks	Awarded for
1	$F \propto I_1 I_2 = I^2$.
1	Factor 9.

Question 5(A) Total: 10 Marks

Question 5 · Option B

Current, resistance and power; a series-parallel circuit — 10 marks

Given: Circuit: $V = 150 \text{ V}$, $R_1 = 12 \Omega$ in series with $R_2 = 6 \Omega$ and $R_3 = 6 \Omega$ in parallel.

5(B)(a)

[3 marks]

Define and give the SI unit of (i) electric current (ii) electrical resistance (iii) electrical power.

ANSWER

$I = Q/t \text{ (A)}; R = V/I \text{ (}\Omega\text{)}; P = IV \text{ (W)}$

WORKING

(i) **Current:** rate of flow of charge, $I = Q/t$. Unit: ampere, A ($= \text{C s}^{-1}$).

(ii) **Resistance:** $R = V/I$, how strongly a conductor opposes the flow of charge. Unit: ohm, Ω .

(iii) **Power:** rate at which electrical energy is used, $P = E/t = IV$. Unit: watt, W.

Marks	Awarded for
1	Current, A.
1	Resistance, Ω .
1	Power, W.

5(B)(b)**[2 marks]**

In $\Delta t = 5 \text{ s}$, $n = 10^{18}$ electrons flow left to right through the area at X. (i) What is the magnitude of the current? (ii) What is the direction of the conventional current?

ANSWER

(i) $I = 0.032 \text{ A}$ (32 mA); (ii) right to left

WORKING

(i) Total charge: $Q = n e = 10^{18} \times 1.602 \times 10^{-19} \text{ C} = 0.160 \text{ C}$.

$$I = Q / \Delta t = 0.160 \text{ C} \div 5 \text{ s} = 0.032 \text{ A}$$

(ii) Conventional current is defined as the direction of flow of **positive** charge, which is opposite to the electrons' motion.

Marks	Awarded for
1	$I = 0.032 \text{ A}$.
1	Right to left (opposite to the electron flow).

5(B)(c)**[5 marks]**

(i) Combined resistance of R_2 and R_3 in parallel? (ii) Total resistance? (iii) Current through R_1 ? (iv) Voltage drop across R_1 ? (v) Total power dissipated?

ANSWER

(i) 3 Ω (ii) 15 Ω (iii) 10 A (iv) 120 V (v) 1500 W

WORKING

(i) Parallel combination:

$$1/R_{23} = 1/6 + 1/6 = 1/3 \Rightarrow R_{23} = 3 \Omega$$

(ii) R_1 is in series with this combination:

$$R_{\text{total}} = 12 + 3 = 15 \Omega$$

(iii) All the current passes through R_1 :

$$I = V / R_{\text{total}} = 150 \div 15 = 10 \text{ A}$$

(iv) and (v):

$$V_1 = I R_1 = 10 \times 12 = 120 \text{ V}; P = V I = 150 \times 10 = 1500 \text{ W}$$

Note on the lecture material: The lecture notes don't cover combining resistors in series or parallel ($R = R_1 + R_2$; $1/R = 1/R_1 + 1/R_2$). These formulas come from outside the lecture material; the $V = IR$ and $P = IV$ steps are from the notes.

Marks	Awarded for
1	(i) 3 Ω .
1	(ii) 15 Ω .
1	(iii) 10 A.
1	(iv) 120 V.
1	(v) 1500 W (own-figure accepted throughout).

Question 5(B) Total: 10 Marks

Question 6 · Option A

Specular and diffuse reflection, refractive index, TIR, the eye as a lens — 10 marks

6(A)(a)

[2 marks]

Explain the difference between specular and diffuse reflection. Draw a simple diagram to illustrate this difference.

ANSWER

Specular: smooth surface, parallel rays reflected together.
Diffuse: rough surface, rays scattered in many directions.

WORKING

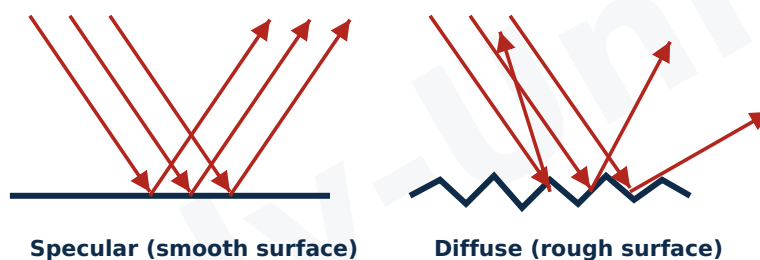


Fig. 3 — Parallel rays stay parallel after reflecting off a smooth surface, but scatter in many directions off a rough one.

Specular reflection happens at a smooth surface (e.g. a mirror). Parallel incident rays are all reflected in the same direction, because the normal is the same everywhere and $\theta_i = \theta_r$. It forms a clear image.

Diffuse reflection happens at a rough surface (e.g. paper). The normal points in different directions at different points, so parallel rays are scattered in many directions and no image is formed. Each ray still obeys $\theta_i = \theta_r$ locally.

Note on the lecture material: Specular vs diffuse reflection is **not covered in the PY1010 lecture notes**, which give only the laws of reflection. This answer is included for completeness.

Marks	Awarded for
1	Correct distinction (smooth vs rough, one direction vs scattered).
1	Diagram.

6(A)(b)(i)**[2 marks]**

A ray exits glass into air. It is incident on the glass-air interface at 20° to the normal and exits at 55.17° . What is the refractive index of the material?

ANSWER

$$n = 2.40$$

WORKING

Snell's law with $n_{\text{air}} = 1$:

$$n \sin 20^\circ = 1 \times \sin 55.17^\circ \Rightarrow n = 0.8208 \div 0.3420 = 2.40$$

Marks	Awarded for
1	Snell's law set up correctly.
1	$n = 2.40$.

6(A)(b)(ii)**[2 marks]**

At what angle of incidence would the light be totally reflected back into the glass at the interface? What is this phenomenon called?

ANSWER

Above $\theta_{\text{crit}} \approx 24.6^\circ$: total internal reflection

WORKING

$$\theta_{\text{crit}} = \sin^{-1}(n_{\text{air}} / n_{\text{glass}}) = \sin^{-1}(1 / 2.40) = 24.6^\circ$$

For any angle of incidence greater than 24.6° , the light is reflected back into the glass.

Marks	Awarded for
1	$\theta_{\text{crit}} \approx 24.6^\circ$ (own-figure accepted).
1	Total internal reflection.

6(A)(c)(i)**[2 marks]**

Treat the eye as a single thin lens 2.5 cm from the retina. If the eye is focused on an object at infinity, what should its focal length be to focus the image on the retina?

ANSWER

$f = 2.5 \text{ cm}$

WORKING

Thin lens equation, $1/u + 1/v = 1/f$, with $u \rightarrow \infty$ so $1/u \rightarrow 0$. The image must form on the retina, so $v = 2.5 \text{ cm}$:

$$1/f = 0 + 1/2.5 \Rightarrow f = 2.5 \text{ cm}$$

Marks	Awarded for
1	$1/u \rightarrow 0$ for an object at infinity.
1	$f = 2.5 \text{ cm}$.

6(A)(c)(ii)**[2 marks]**

If the object is brought to 25 cm from the lens (the near point), what does the new focal length need to be to keep the object in focus?

ANSWER

$f \approx 2.3$ cm (2.27 cm): the lens must become more strongly converging

WORKING

Now $u = 25$ cm and $v = 2.5$ cm (the retina doesn't move):

$$1/f = 1/25 + 1/2.5 = 0.04 + 0.40 = 0.44 \text{ cm}^{-1} \Rightarrow f = 2.27 \text{ cm}$$

Marks	Awarded for
1	Thin lens equation with $u = 25$ cm, $v = 2.5$ cm.
1	$f \approx 2.27$ cm.

Question 6(A) Total: 10 Marks

Question 6 · Option B

Types of wave, $v = f\lambda$, interference of water waves from two stones — 10 marks

Given: Stones land 4 m in front of you and 4 m out but 3 m to the right; wave period 0.8 s; crest spacing 2 m.

6(B)(a)

[2 marks]

Explain the difference between transverse and longitudinal waves, with an example of each.

ANSWER

Transverse $\perp$ to propagation (light); longitudinal $\parallel$ to propagation (sound)

WORKING

Transverse: the medium moves **perpendicular** to the direction the wave travels, e.g. light, or water waves.

Longitudinal: the medium vibrates **parallel** to the direction the wave travels (compressions and rarefactions), e.g. sound.

Marks	Awarded for
1	Correct distinction.
1	One example of each.

6(B)(b)**[2 marks]**

Orchestras tune to A at 440 Hz. With a speed of sound of 340 m/s, what is the wavelength of this note in air?

ANSWER

$$\lambda \approx 0.77 \text{ m}$$

WORKING

$$\lambda = v / f = 340 \text{ m s}^{-1} \div 440 \text{ Hz} = 0.77 \text{ m}$$

Marks	Awarded for
1	$\lambda = v/f.$
1	0.77 m.

6(B)(c)(i)-(ii)**[3 marks]**

(i) Calculate the frequency of the wave in SI units. (ii) Calculate the velocity of the wave.

ANSWER

(i) $f = 1.25 \text{ Hz}$ (ii) $v = 2.5 \text{ m s}^{-1}$

WORKING

(i) The period is $P = 0.8 \text{ s}$, so:

$$f = 1 / P = 1 \div 0.8 \text{ s} = 1.25 \text{ Hz}$$

(ii) The distance between successive crests is the wavelength, $\lambda = 2 \text{ m}$:

$$v = f \lambda = 1.25 \text{ Hz} \times 2 \text{ m} = 2.5 \text{ m s}^{-1}$$

Marks	Awarded for
1	$f = 1/P = 1.25 \text{ Hz}$.
1	$\lambda = 2 \text{ m}$ identified.
1	$v = 2.5 \text{ m s}^{-1}$.

6(B)(c)(iii)-(iv)**[3 marks]**

(iii) Calculate the direct distance d from where the second stone lands to where you are standing. (iv) When the first waves from each stone meet in front of you, will a large wave or no wave be produced? Explain.

ANSWER

(iii) $d = 5 \text{ m}$ (iv) Path difference = $\lambda/2$, so destructive interference: no wave

WORKING

(iii) The two sides (4 m and 3 m) are at right angles, so by Pythagoras:

$$d = \sqrt{4^2 + 3^2} = 5 \text{ m}$$

(iv) Both stones land at the same time, so the waves start in phase. The path difference to you is:

$$\Delta = 5 \text{ m} - 4 \text{ m} = 1 \text{ m} = \lambda/2$$

Half a wavelength means the waves arrive **exactly out of phase**: crest meets trough and they cancel (destructive interference).

Note on the lecture material: The notes show that waves exactly out of phase cancel, but don't state the path-difference condition $(n + \frac{1}{2})\lambda$ explicitly. The reasoning here combines the lectures' superposition pages with λ as the crest spacing.

Marks	Awarded for
1	$d = 5 \text{ m}$.
1	Path difference = $1 \text{ m} = \lambda/2$.
1	Destructive: no wave.

Question 6(B) Total: 10 Marks